

AI Assignment 1: Agent Framework and Environment Analysis

Student Name: Jaswant Kumar

Roll Number: 2K23/CSE/61

Topic: IoT-based Soil Monitoring and Crop Recommendation System (**AgriMind**)

1. Task Description

For this assignment, I have formulated a framework for an **IoT-based Soil Monitoring System (AgriMind)**. Inspired by my personal background in rural agriculture near Naukot, Sindh, this system aims to modernize traditional farming practices. It utilizes a sensor-integrated probe to collect real-time soil data and an AI agent to provide farmers with data-driven crop recommendations and fertilizer suggestions.

2. PEAS Specification

The system is defined using the PEAS framework to ensure all engineering requirements are met before implementation:

- **Performance Measure:** Accuracy of nutrient detection (NPK), percentage increase in crop yield, reduction in fertilizer wastage, and system response time.
- **Environment:** Agricultural fields in Sindh with diverse soil types (sandy, clay), varying weather conditions, and intermittent internet connectivity.
- **Actuators:** Mobile application interface for displaying diagnostic reports, notification systems for fertilizer alerts, and cloud-based data storage.
- **Sensors:** 5-in-1 NPK sensor probe (measuring Nitrogen, Phosphorus, Potassium, pH, and moisture levels) and user input regarding historical crop data.

3. Task Environment Classification

The AgriMind environment is classified across the following dimensions to determine algorithmic difficulty:

- **Partially Observable:** The internal chemical composition of the soil is not visible unless the sensor probe is physically inserted.
- **Single-Agent:** The individual probe operates independently to analyze its specific plot of land.

- **Stochastic:** Changes in soil nutrients due to sudden weather shifts or pest activities are unpredictable.
- **Sequential:** Decisions made regarding current fertilizer application will impact future soil health and subsequent readings.
- **Dynamic:** The environment (moisture and temperature) can change while the agent is processing a recommendation.
- **Continuous:** Nutrient values and pH levels are represented in a continuous range rather than discrete steps.

4. Critical Analysis

Greatest Technical Challenge

The primary challenge in designing AgriMind is managing a Stochastic and Partially Observable environment. Unlike a board game, agricultural variables like soil chemistry and weather are unpredictable. The AI must be robust enough to handle "noisy" sensor data and provide accurate recommendations even when certain environmental factors remain hidden or uncertain.

Utility-Based Agent and Trade-offs

A **Utility-Based Agent** structure is chosen because it allows the system to balance conflicting goals through a utility function.

- **Utility Function:** $U = w_1(\text{Accuracy}) + w_2(\text{Battery Efficiency})$.
- **Trade-off:** High-frequency data scanning improves accuracy but depletes the probe's battery rapidly.
- **Behavioral Change:** If the weight of **Battery Efficiency (w_2)** is doubled, the agent will prioritize longevity. It will reduce the frequency of soil scans, opting to provide slightly older data to ensure the hardware remains operational for a longer duration without maintenance.